Common theory for projective markings and rotations

The human_compression project

11 October 2026

Abstract

Projective centralizers and matrix intertwiners control both involutory string markings and rotational markings. We retain the complete common proof for the three existing polytope and maniplex applications, then prove simultaneous reversal, a one-parameter classification of two affine ends, and one trace quadratic for the three exceptional rank-four families. An anchored-involution interface supplies the chiral rank bound; the previously accepted norm-polynomial and translated-subfield arguments supply the odd-composite-degree construction. Finite presentations and small-field exceptions are paid by complete certificates. Classical subgroup, automorphism, character-sum and geometric correspondence inputs remain explicit. Five source applications use this whole common paper.

1 Markings and retained inputs

A *string marking* of a group G is a generating ordered tuple $(r_0, \dots, r_{n-1})$ of distinct nonidentity involutions with $r_i r_j = r_j r_i$ whenever $|i-j| > 1$. Put $G_I = \langle r_i : i \in I \rangle$, with $G_\varnothing = 1$. Its labels are $|r_0 r_1|, \dots, |r_{n-2} r_{n-1}|$. The marking is a string C-marking if, in addition,

$$G_I \cap G_J = G_{I \cap J} \qquad (I, J \subseteq \{0, \dots, n-1\}). \tag{1}$$

For rank four write $H = \langle r_0, r_1, r_2 \rangle$ and $J = \langle r_1, r_2, r_3 \rangle$ for its two ends. Throughout, $L(q) = \mathrm{PSL}_2(q)$ and $P(q) = \mathrm{PGL}_2(q)$, for a prime power q.

We retain the following classical inputs, in exactly the forms used below. Dickson's classification says that a proper subgroup of $L(q)$ or $P(q)$, for odd q, is cyclic, dihedral, elementary abelian of characteristic order, a subgroup of an affine group $V \rtimes k^\times$, one of A_4, S_4, A_5, or a conjugate of a standard subfield group $L(k)$ or $P(k)$. Only groups which actually embed are intended. In the subfield alternatives $k \subseteq \mathbb{F}_q$ and the embedding is the natural matrix embedding, not an arbitrary abstract group isomorphism. The possible same-field proper subgroup $L(q) < P(q)$ must be retained. We use

$$|L(q)| = \frac{q(q^2-1)}{(2, q-1)}, \qquad |P(q)| = q(q^2-1),$$

the simplicity of $L(q)$ for $q \geq 4$, and $C_{P(q)}(L(q)) = 1$. The small identifications are $L(2) = P(2) \cong S_3$, $L(3) \cong A_4$, $P(3) \cong S_4$, $L(5) \cong A_5$ and $P(5) \cong S_5$. These subgroup and structural facts are classical; subgroup multiplicity formulas from the sources are not inputs.

We also retain the correspondence between ordered string C-markings and regular abstract polytopes, including the following clauses. Rank-i faces are cosets of $G_{\{0,\dots,n-1\}\setminus\{i\}}$; reversal gives duality; a marking-preserving epimorphism of string C-groups induces a covering. The local markings $[3,5]_5$ and $[5,3]_5$ below are the usual antipodal icosahedral and dodecahedral quotients. These standard local-map names and the simplex interpretation are retained geometric identifications. They do not replace the required verification of the actual markings.

All groups $L(q), P(q)$ are centre-free and directly indecomposable. For the simple groups this is immediate. To see the assertion for $P(q)$ when $q \geq 4$, let commuting subgroups A, B generate $P(q)$;

both are normal. The subgroup $[L(q), A]$ lies in $L(q) \cap A$ and is normal in the simple group $L(q)$. If it is nontrivial, $L(q) \leq A$, so B centralizes $L(q)$ and is trivial. Otherwise A centralizes $L(q)$ and is trivial. The small cases follow from the normal-subgroup lists $1, A_3, S_3$; $1, V_4, A_4$; and $1, V_4, A_4, S_4$. In any string marking of these groups every adjacent product has order at least three. Indeed, a commuting adjacent pair splits the generators into two nontrivial commuting generated subgroups. Their intersection is central, hence trivial, and the resulting direct product contradicts indecomposability. This argument does not assume (1).

2 Common centralizers

Lemma 1. *In characteristic two, the centralizer of a nonidentity involution in $P(q)$ is abelian. Let $k \subseteq K$ be finite fields of odd characteristic, and let $a, b \in P(k)$ be involutions with $|ab| > 2$. Their common centralizer in $P(K)$ lies in $P(k)$. It is trivial when ab is unipotent and has order two otherwise. In the latter case its nonidentity element is represented by the invertible matrix*

$$Z = A - \tfrac{1}{2}\operatorname{tr}(A)I, \tag{2}$$

where $A \in \mathrm{GL}_2(k)$ represents ab.

Proof. In characteristic two an involution is unipotent. Over an algebraic closure conjugate it to $x \mapsto x+1$. Its unique fixed point is infinity, so a commuting projectivity is $x \mapsto cx+d$. Commutation gives $c = 1$; these translations form an abelian group. The centralizer over the original field is a subgroup of this group.

In odd characteristic first suppose $u = ab$ is semisimple. Over the algebraic closure write $u : x \mapsto \lambda x$, with $\lambda \neq \pm 1$. A commuting projectivity preserves each of its two fixed points, since interchanging them would send u to $u^{-1} \neq u$. It is therefore $x \mapsto cx$. As $aua^{-1} = u^{-1}$, the involution a has the form $x \mapsto d/x$. Commutation with a gives $c^2 = 1$, so the common centralizer is $\{1, x \mapsto -x\}$. If A has eigenvalues α, β, the matrix in (2) has the nonzero eigenvalues $(\alpha - \beta)/2$ and $(\beta - \alpha)/2$. It represents $x \mapsto -x$ and has entries in k. Rescaling A only rescales Z, proving the claimed descent, including the nonsplit case.

If u is unipotent, conjugate it to $x \mapsto x+1$. Its centralizer again consists of translations $x \mapsto x+c$. An involution conjugating it to its inverse has the form $x \mapsto -x+d$. A commuting translation satisfies $2c = 0$, and hence $c = 0$. These cases exhaust u. □

Lemma 2. *Let G be centre-free with a string marking. Its first two generators do not commute. If every noncommuting pair of involutions has at most one nonidentity common centralizer element, the rank is at most four. If every involution centralizer is abelian, the rank is at most three.*

Proof. If r_0 commuted with r_1, it would commute with every generator by the string relations, so would be a nonidentity central element. At rank at least five, both r_3 and r_4 centralize r_0, r_1, contradicting the first centralizer hypothesis and their distinctness. Under the second hypothesis a rank-four marking puts r_0, r_1 in the abelian centralizer of r_3, again a contradiction. □

Lemma 3. *Let a rank-four string marking lie in $P(q)$, with q odd and adjacent labels greater than two. If H is proper, then $|r_0 r_1|$ is odd; if J is proper, then $|r_2 r_3|$ is odd.*

Proof. The nonidentity element r_3 centralizes $D = \langle r_0, r_1 \rangle$. Lemma 1 excludes the unipotent case and gives a unique nonidentity common centralizer element. If $m = |r_0 r_1|$ were even, $(r_0 r_1)^{m/2}$ would be that element. Then $r_3 \in H$, contrary to $G = \langle H, r_3 \rangle$ and properness. Reverse the indices for J. □

Theorem 4 (Projective reduction)**.** *A string marking of $L(q)$ or $P(q)$ has rank at most four; rank four requires odd q. In rank four every proper end of $L(q)$ is S_4 or A_5. Every proper end of $P(q)$ is S_4, A_5, or its natural subgroup $L(q)$. If both ends are proper, both are S_4 or A_5 and both outer labels are odd. These statements require no intersection property.*

Proof. The rank claims follow from Lemmas 1 and 2, since the projective groups are centre-free. This also handles $q = 2$; $L(3)$ has no generating tuple of involutions, since its involutions generate only V_4. All adjacent labels in the remaining cases exceed two by the first section.

Let $H = \langle r_0, r_1, r_2 \rangle$ be proper and put $D = \langle r_0, r_1 \rangle$. A cyclic or elementary abelian group cannot contain the nonabelian group D. Involutions in A_4 generate only V_4. In an affine group in odd characteristic every involution is $x \mapsto -x + v$; two such involutions commute only when they coincide. Thus the distinct commuting endpoints r_0, r_2 exclude this alternative, including affine dihedral groups of order $2p$.

If H is dihedral, its distinct commuting involutions r_0, r_2 force its rotation subgroup to have even order. (In an odd dihedral group two reflections commute only when equal.) Its central involution z and r_3 both centralize D. Lemma 1 gives $r_3 = z$, so $r_3 \in H$, a contradiction. This exclusion uses properness, not an intersection identity inside H.

If H is a natural subfield group, conjugate it into $P(k)$. Lemma 1 puts r_3 in the same $P(k)$, hence puts all of G there. For a proper subfield of size s we have $q \geq s^2$, and

$$|G| \geq q(q^2 - 1)/2 > s(s^2 - 1) = |P(k)|,$$

a contradiction. The only remaining same-field proper alternative is $H = L(q) < P(q)$. This proves the asserted lists of proper ends. The qualification "natural" is essential: exceptional $A_5 \cong L(5)$ and $S_4 \cong P(3)$ in other characteristics are not subfield exclusions.

Suppose now that both ends are proper. Lemma 3 makes both outer labels odd. If $H = L(q) < P(q)$, then H is normal of index two; $r_2 \in H$ and $r_3 \notin H$. The image of $r_2 r_3$ in $P(q)/H$ therefore has order two, impossible for its odd order. Reversal excludes the same alternative for J. The remaining ends are S_4, A_5. □

3 One finite theorem for exceptional ends

For a triple $f = (m, n, k)$ write

$$[m, n]_k = \langle x, y, z \mid x^2 = y^2 = z^2 = (xz)^2 = (xy)^m = (yz)^n = (xyz)^k = 1 \rangle. \tag{3}$$

The third parameter is the Petrie order, not an adjacent label. The six ordered local classes are

$$\begin{array}{c|ccc} S_4 & (3,3,4) & (3,4,3) & (4,3,3) \\ A_5 & (3,5,5) & (5,3,5) & (5,5,3). \end{array} \tag{4}$$

For $f = (m, n, k)$ and $v = (n, s, l)$ in this list define

$$\begin{aligned} U(f, v) = \langle r_0, r_1, r_2, r_3 \mid\ & r_i^2 = 1,\ (r_i r_j)^2 = 1\ (j - i > 1), \\ & (r_0 r_1)^m = (r_1 r_2)^n = (r_2 r_3)^s = 1, \\ & (r_0 r_1 r_2)^k = (r_1 r_2 r_3)^l = 1 \rangle. \end{aligned} \tag{5}$$

Matching the middle label gives fourteen ordered pairs. Reversal sends (f, v) to (v^*, f^*), where $(a, b, c)^* = (b, a, c)$. There are ten reversal orbits, six fixed. Their complete table is:

Row	f	v	Order	Group	C-marking	Outer labels
1	$(5,5,3)$	$(5,5,3)$	1	trivial	no	5, 5
2	$(5,5,3)$	$(5,3,5)$	1	trivial	no	5, 3
3	$(5,3,5)$	$(3,5,5)$	3420	$L(19)$	yes	5, 5
4	$(5,3,5)$	$(3,4,3)$	60	A_5	no	5, 4
5	$(5,3,5)$	$(3,3,4)$	1	trivial	no	5, 3
6	$(4,3,3)$	$(3,4,3)$	96	$2^4 : S_3$	yes	4, 4
7	$(4,3,3)$	$(3,3,4)$	24	S_4	no	4, 3
8	$(3,5,5)$	$(5,3,5)$	660	$L(11)$	yes	3, 3
9	$(3,4,3)$	$(4,3,3)$	1	trivial	no	3, 3
10	$(3,3,4)$	$(3,3,4)$	120	S_5	yes	3, 3

Here an entry "yes" means that the specified generators, not merely some marking of the abstract group, form a string C-group. Rows which collapse do not have the indicated local groups; the local triples label their presentation input. The order-96 row has adjacent outer labels $4, 4$, not $3, 3$.

Lemma 5 (Exact finite certificate)**.** *Every rank-three string marking of S_4 or A_5 has the intersection property. Up to the full automorphism group its ordered type and Petrie order are exactly* (4)*;* (3) *is an exact presentation in each case. The ten-row table above is correct, including its groups, specified markings, collapse assertions and reversal coverage. In rows* $3, 8, 10$*, reversal extends to an automorphism of the marked group.*

Proof. The complete program `required_finite.g` enumerates all involutions in S_4, A_5, all ordered pairs of distinct commuting endpoints x, z, and every middle involution y. Its admission tests are only distinctness and generation. It admits 72 and 360 triples respectively. For every admitted triple it then checks both adjacent orders and all 64 subset-pair identities in (1). Thus IP is an output, never a filter whose use would assume the desired category change. The full automorphism orbits give the three classes in each row of (4). The code constructs each literal presentation (3) and computes its order 24 or 60. The marking provides a surjection onto the corresponding finite group; equality of orders makes it an isomorphism.

The code independently generates all fourteen compatible pairs, partitions by reversal and checks the ten supplied representatives and six fixed pairs. For each literal presentation (5), it computes the finite presentation order, constructs an isomorphic permutation group and transports the four original generators. It checks generator orders and distinctness, then tests IP over all sixteen subsets. Every accepted marking satisfies all 256 subset-pair identities; a rejected row may fail a generator condition or terminate at its first failed intersection. Thus the negative entries certify failure of the complete string C-marking conditions, not necessarily a separate IP test. Rows $3, 8, 10$ are identified by actual group isomorphisms with $L(19), L(11), S_5$; generator reversal is checked by constructing the homomorphism and testing bijectivity. In rows $4, 7$ it verifies the indicated group and that one end is the whole group. In row 6 it verifies an elementary abelian normal subgroup of order 16, quotient S_3 and a complement. The four trivial rows are checked directly. These exact computations prove the finite assertions relative to GAP's retained finite-group and finite- presentation algorithms; they are not claimed as Lean kernel computations. □

Theorem 6 (Exceptional ends and odd outer labels)**.** *Let $(G; r_0, r_1, r_2, r_3)$ be a string marking whose two ends are proper and isomorphic to S_4 or A_5. If both outer labels are odd, then, up to isomorphism preserving the ordered marking, it is exactly one of rows* $3, 8, 10$*. Each row supplies one class, satisfies IP, and has type* $\{5, 3, 5\}$*,* $\{3, 5, 3\}$ *or* $\{3, 3, 3\}$ *respectively. In particular the statement does not require G to be a projective group.*

Proof. The local census and its exact presentations give a marking-preserving surjection from one of the ten groups $U(f, v)$ to G, after possibly reversing the marking. It is necessary to examine quotients, not just the universal groups themselves. The trivial rows cannot give nonidentity markers. In rows $4, 7$, an end already equals the full universal group; this equality survives every quotient, contradicting properness. Row 6 has first and last local product orders four, whereas the given ends have their exact local markings and hence these exact orders. It is excluded by oddness.

The remaining universals are the simple groups $L(19), L(11)$ and S_5. The simple groups have no nontrivial proper quotient. The normal subgroups of S_5 are $1, A_5, S_5$, so its only nontrivial proper quotient is C_2, which cannot have an end S_4 or A_5. Thus every surviving map is an isomorphism on all four distinguished generators. Lemma 5 gives IP and the asserted types. The local classes were taken under full automorphisms, so no marking was lost through a conjugacy restriction. The reversal automorphisms verified below identify the two representatives of each reversal orbit. There is consequently one ordered-marking class per surviving row. □

Sharpness of the parity hypothesis. The verified order-96 row has proper S_4 ends, type $\{4, 3, 4\}$ and IP. It therefore contradicts the three-case conclusion if the odd-outer-label hypothesis is removed. We make no classification of all quotients without that hypothesis.

Corollary 7. *Every rank-at-least-four string marking of $L(q)$ or $P(q)$ with both ends proper is one of these three markings. The first two occur in $L(19), L(11)$ respectively; the third occurs in $P(5)$. A string C-marking automatically has both ends proper, so the same conclusion applies to all projective string C-markings of rank at least four.*

Proof. Theorem 4 supplies rank four, the exceptional ends and odd outer labels; apply Theorem 6. The order formulas are strictly increasing in odd q. The groups $L(11), L(19)$ are simple, whereas odd-characteristic $P(q)$ has a proper subgroup of index two. The group S_5 is not simple and has order $120 = |P(5)|$; the small $L(3)$ is not S_5. These observations give exactly the stated field values and projective types. Finally, if an omitted generator belonged to its complementary subgroup, (1) would intersect that subgroup with its generator subgroup nontrivially. Hence C-markings have proper ends. □

4 Geometry and universal coverings

Theorem 8. *The three markings in Theorem 6 give the 57-cell, the 11-cell and the 4-simplex. Their types, local maps and face vectors are*

group	*type*	*facet / vertex figure*	(f_0, f_1, f_2, f_3)
$L(19)$	$\{5, 3, 5\}$	*hemi-dodecahedron / hemi-icosahedron*	$(57, 171, 171, 57)$
$L(11)$	$\{3, 5, 3\}$	*hemi-icosahedron / hemi-dodecahedron*	$(11, 55, 55, 11)$
S_5	$\{3, 3, 3\}$	*tetrahedron / tetrahedron*	$(5, 10, 10, 5)$.

All three are self-dual and universal among regular polytopes with their respective facet and vertex-figure types. The 11-cell edge graph is K_{11}. The even-word subgroup is the full group in the two simple rows, and is A_5 in the simplex row.

Proof. The two simple rows have local presentations $[3, 5]_5, [5, 3]_5$, with the retained antipodal map interpretations. Their exact global presentations are the string Coxeter relations of types $\{3, 5, 3\}$ and $\{5, 3, 5\}$, together with

$$(r_0 r_1 r_2)^5 = (r_1 r_2 r_3)^5 = 1.$$

For the third row send the four generators to $(12), (23), (34), (45)$. These generate S_5 and satisfy every relator in (5), including the two fourth powers of consecutive triple products. Both groups have order 120, so this is an isomorphism of marked groups with the standard simplex marking.

The program verifies that generator reversal extends to an automorphism in each of the three rows; the retained correspondence gives self-duality. It also computes each of the four complementary-subgroup indices, giving the displayed face vectors. In the simple rows this has the following direct interpretation. Both ends have order 60. An edge stabilizer is $\langle r_0 \rangle \times \langle r_2, r_3 \rangle$, since the factors commute and IP makes their intersection trivial. Its order is $4m$ for outer label $m = 3$ or 5; the two-face stabilizer has the same order. Thus $660/60 = 11$, $660/12 = 55$, $3420/60 = 57$ and $3420/20 = 171$ give the first two vectors. For the 11-cell the code constructs actual vertex and edge cosets and checks their incidences: every edge has two endpoints and the 55 resulting unordered pairs are distinct. There are exactly 55 pairs on eleven vertices, proving K_{11} rather than inferring it merely from face counts.

The even-word subgroup is normal of index at most two: in its quotient all involutions have the same image. Simplicity makes it the full group in the two simple rows. In the simplex marking parity is the ordinary sign homomorphism, giving A_5.

Let Q be any regular polytope with one of the three indicated pairs of local maps. Its distinguished generators satisfy the corresponding literal presentation (5). There is therefore a marked epimorphism from the already verified string C-group of that row to its automorphism group. The retained covering

theorem gives a covering of Q by the corresponding polytope above. No finiteness assumption on Q is needed: the epimorphism itself forces its group to be finite. This is universality of these exact markings, not an assumption that every abstract amalgam has IP. □

5 The common extension construction

We use the standard torus structure in $L(q)$ for odd q: its split and nonsplit cyclic tori have orders $(q-1)/2, (q+1)/2$, with dihedral normalizers of twice these orders. The split torus consists of diagonal projectivities of square eigenvalue ratio. The nonsplit torus comes from multiplication on $\mathbb{F}_{q^2}$ modulo scalars, with the determinant- square condition. Reflections in the normalizer interchange the two eigenlines. These classical torus facts are retained; no maximality of these normalizers is assumed.

Theorem 9 (Half-turn extension)**.** *Let q be odd, choose $\alpha \in \{1, -1\}$ with $q \equiv \alpha \pmod 4$, and put $N = (q-\alpha)/2$. Let (a, b, c) be a rank-three string marking of $H \leq L(q)$, with $|bc| > 2$ and $|ab| = m > 1$ an odd divisor of N. There is a unique nonidentity projectivity z centralizing a, b; it lies in $L(q)$ and (a, b, c, z) is a string marking of $\langle H, z\rangle$. If $H = L(q)$, this group remains $L(q)$. If H is maximal proper in $L(q)$ and $C_H(\langle a, b\rangle) = 1$, it is again $L(q)$ and its first end is exactly H.*

Proof. The element ab is semisimple, since m divides $q - \alpha$ and hence is prime to the characteristic. It has two eigenlines over at most a quadratic extension. The two torus orders are coprime consecutive integers; since $m > 1$ divides N, its torus has order N. Both a, b invert ab, so preserve its unordered eigenline pair and normalize this torus. The even cyclic torus has a unique involution z. It commutes with the torus and its inverting normalizer coset, so centralizes a, b. Lemma 1 proves its asserted uniqueness even in $P(q)$.

The dihedral group $\langle a, b\rangle$ has odd rotation order and trivial centre. Thus z does not belong to it and differs from a, b. It also differs from c, because it commutes with b whereas $|bc| > 2$. The four generators are distinct nonidentity involutions and have every required distant commutation. They generate $\langle H, z\rangle$ by construction. If $H = L(q)$ there is nothing further to prove. Under the second hypothesis $z \notin H$, because its membership would give a nonidentity element of the stated trivial internal centralizer. Maximality then gives $\langle H, z\rangle = L(q)$. □

The extension theorem concerns string markings; it does not assert IP for the extended object. The maniplex application proves its category correspondence and hence the assertion that the first end describes the original map. Its all-prime constructions and explicit source-basis identifications are local proofs in that application, not finite-sample conclusions.

6 Rotations and their retained geometric interface

For the new applications we retain the rotation-group correspondence of Schulte–Weiss [6]. A rank-n rotation marking is an ordered generating tuple $(\sigma_1, \dots, \sigma_{n-1})$ with every interval product $\sigma_i \cdots \sigma_j$, $i < j$, a nonidentity involution. For a set I of vertex indices put

$$G_I^+ = \langle \sigma_i \cdots \sigma_j \,:\, i \leq j,\ i-1, j \in I\rangle.$$

The full rotation intersection property is $G_I^+ \cap G_J^+ = G_{I\cap J}^+$. In rank four it is equivalent to the two adjacent cyclic intersections being trivial and the two ends intersecting in the middle cyclic group. Such a marking gives a chiral polytope exactly when it has no involutory automorphism sending σ_1 to σ_1^{-1}, σ_2 to $\sigma_1^2\sigma_2$ and fixing the later rotations. Otherwise it gives a directly regular polytope. This correspondence, including the equivalence of intersection conditions, is retained rather than claimed as a new theorem.

We also retain the full semilinear automorphism theorem $\mathrm{Aut}(L(q)) = \mathrm{P\Gamma L}_2(q)$ for $q \geq 4$ and its corresponding $P(q)$ form, with the standard small identifications. Finite-field norm and cyclic-character facts and the exact multiplicative character bound from [8] are the only new analytic inputs. The projective subgroup alternatives are the same classical inputs as in the first section, used in their natural-subfield form.

7 One simultaneous reversal for two projectivities

Theorem 10. *Let F be a field and let $A, B \in \mathrm{GL}_2(F)$ have no common invariant line over an algebraic closure. Then $X = AB - BA$ is invertible, X^2 is scalar, and conjugation by $[X]$ simultaneously inverts $[A], [B] \in \mathrm{PGL}_2(F)$. In particular, every generating pair of $\mathrm{PSL}_2(q)$ or $\mathrm{PGL}_2(q)$ admits an automorphism of order dividing two which inverts both generators.*

Proof. The Cayley–Hamilton identity gives

$$XA + AX = \mathrm{tr}(A)X, \qquad XB + BX = \mathrm{tr}(B)X.$$

Indeed, the first left side is $A^2B - BA^2$, and the second is $AB^2 - B^2A$. Thus $XA = \mathrm{adj}(A)X$ and $XB = \mathrm{adj}(B)X$. These identities hold in every characteristic. Also $\mathrm{tr}(X) = 0$, whence $X^2 = -\det(X)I$.

To see that X is invertible, extend scalars to an algebraic closure. Neither A nor B is scalar, since otherwise they share an eigenline. If $A = \mathrm{diag}(a, d)$ with $a \neq d$ and $B = \left(\begin{smallmatrix} u & v \\ w & z \end{smallmatrix}\right)$, then $\det(AB - BA) = (a - d)^2 vw$. Its vanishing makes either coordinate line common and invariant. Otherwise A has one Jordan block and, after rescaling its off-diagonal entry, the commutator is $\left(\begin{smallmatrix} w & z-u \\ 0 & -w \end{smallmatrix}\right)$. Singularity again makes the unique eigenline of A invariant under B. Both possibilities contradict the hypothesis. Since $\mathrm{adj}(A) = \det(A)A^{-1}$, the two matrix identities now prove the projective assertion.

For the finite projective groups, the upper and lower elementary unipotents have distinct unique fixed points over the algebraic closure. Both belong to $\mathrm{PSL}_2(q)$. Hence a pair generating either projective group cannot share an invariant line. Conjugation by $[X]$ normalizes $\mathrm{PGL}_2(q)$ and its determinant-square subgroup $\mathrm{PSL}_2(q)$, and its square is the identity. □

For rank-three rotations a, b, apply the theorem to the generating pair a, ab. Since ab is an involution, the resulting involutory automorphism sends a to a^{-1} and fixes ab, hence sends b to a^2b. This is exactly the rank-three rotation reversal rule. The retained rotation-group correspondence therefore excludes a chiral polyhedron with either projective group as its full automorphism group. The correspondence is the retained geometric input; the preceding theorem supplies the complete algebraic reversal. It does not require the product of the two generators to be an involution or their cyclic subgroups to have trivial intersection. This recovers Elliott–Leemans–O'Reilly-Regueiro, Theorem 4.1, by one argument, including the small fields.

For clarity, the source's auxiliary involution count requires a characteristic qualification. For $u : x \mapsto x + 1$, an inverting projectivity fixes infinity and has the form $x \mapsto ax + b$. The conjugation equation forces $a = -1$. In odd characteristic all q choices of b give nonidentity involutions. In characteristic two these are translations; the choice $b = 0$ is the identity and the other $q - 1$ choices are involutions. Thus the literal count in its Lemma 4.3 is $q - 1$ when the characteristic is two. This correction does not affect the principal nonexistence theorem.

8 Two affine ends and one field parameter

Put $L(q) = \mathrm{PSL}_2(q)$ and $P(q) = \mathrm{PGL}_2(q)$. We retain the standard theorem that their automorphisms are induced by projectivities and field automorphisms (with the usual small identifications). No subgroup-classification theorem is needed in the following construction.

Theorem 11. *Let $F = \mathbb{F}_q$, let $t \in F \setminus \{0, 1, -1\}$ generate F as a field over its prime field, and put $k = |t|$. Consider the projective matrices*

$$A = \begin{pmatrix} -t^{-1} & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} t & 0 \\ 0 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 \\ 1+t & -t \end{pmatrix}.$$

Their pair products AB, BC and triple product ABC are nonidentity involutions. Set $D = \langle -1, t \rangle \leq F^\times$. The groups $\langle A, B \rangle$ and $\langle B, C \rangle$ are the affine groups $F^+ \rtimes D$ fixing infinity and zero respectively; their intersection is the diagonal group D. Consequently the rank-four rotation intersection conditions hold exactly when $-1 \in \langle t \rangle$. In that case $\langle A, B, C \rangle$ equals $L(q)$ if t is a square, and $P(q)$ otherwise; in characteristic two these groups coincide. The resulting polytope has type

$$(|-t|, k, |-t|).$$

It is chiral exactly when no field automorphism sends t to t^{-1}. For a fixed admissible k, the number of isomorphism classes of the ordered chiral markings is $\varphi(k)/[F : \mathbb{F}_p]$.

Proof. The displayed products have trace zero and are nonscalar; direct multiplication proves the three involution relations in every characteristic. The first end fixes infinity. It contains B and $R = AB = \left(\begin{smallmatrix} -1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. The product $RBRB^{-1}$ is translation by $1 - t$. Conjugation by B scales translations by t, so its translation subgroup contains $(1-t)\mathbb{F}_p[t] = F$. Its group of slopes is precisely D. This proves the first affine description. After interchanging zero and infinity, the second end contains a diagonal of slope t^{-1} and an affine map of slope $-t$ and translation $1 + t$. The same commutator argument supplies all translations, since $t \neq \pm 1$, and proves its description. An element in both ends fixes zero and infinity, so is diagonal; both ends contain exactly the slopes D there.

The nonidentity powers of A fix precisely infinity and $t/(1 + t)$; those of B fix infinity and zero. Thus their cyclic intersection is trivial by the uniqueness of a projectivity on three distinct points. Likewise $\langle B \rangle \cap \langle C \rangle = 1$, since the fixed points of C are zero and one. Together with the computed end intersection this proves precisely the rank-four intersection conditions.

The two ends contain the full opposite elementary unipotent groups. These generate $L(q)$: elementary row operations reduce any determinant-one matrix to a diagonal matrix, and such a diagonal is also a product of upper and lower elementary matrices. The determinants of A, B, C have classes $[-t^{-1}], [t], [-t]$. Under the intersection hypothesis, $-1 \in \langle t \rangle$, their classes generate just $[t]$. This proves the claimed whole group, without a maximal-subgroup test. The eigenvalue ratios give the type. In odd characteristic the intersection hypothesis says k is even; then $|-t| = k$ for $4 \mid k$ and $|-t| = k/2$ for $k \equiv 2 \pmod 4$. In characteristic two it is k.

For chirality use the retained rank-four rotation criterion: a reversing automorphism would send A to A^{-1}, B to A^2B and fix C. Write it as a projectivity after a field automorphism σ. Since C is semisimple with eigenvalue ratio $-t$ and has order greater than two, fixing its projective conjugacy class forces $t^\sigma \in \{t, t^{-1}\}$. If $t^\sigma = t$, the full-field hypothesis makes σ the identity. A projectivity centralizing C then fixes zero and one individually, whereas one inverting A must interchange infinity and $u = t/(1 + t)$. There is no such projectivity: writing it as $\left(\begin{smallmatrix} a & 0 \\ c & d \end{smallmatrix}\right)$, these conditions give $a = c + d$, $a = uc$ and $d = -uc = -a$. Thus $c = 2a$ and $u = 1/2$, forcing $t = 1$; in characteristic two they already give $c = 0$, contradicting the interchange. This proves impossibility of inner reversal.

If instead $t^\sigma = t^{-1}$, then $\sigma^2 = 1$ because t generates F. The matrix R above is defined over the prime field and satisfies $R^2 = I$. Direct substitution gives

$$RA^\sigma R = A^{-1}, \qquad RB^\sigma R = A^2B, \qquad RC^\sigma R = -t^{-1}C.$$

Thus $X \mapsto RX^\sigma R$ is the required involutory projective automorphism, and the polytope is directly regular. This establishes both directions of the chirality assertion.

Finally an isomorphism of two ordered markings sends the first end to the first end and the second to the second. Their unique fixed points are infinity and zero, so the projective part of its semilinear

action fixes these individually. It is diagonal, and comparison of B shows that the new parameter is t^σ. Conversely field automorphisms send the whole displayed triple to the triple with parameter t^σ. Every such orbit has length $[F : \mathbb{F}_p]$, and there are $\varphi(k)$ elements of order k. This gives the stated number. □

Proposition 12. *Suppose a rank-four rotation marking generates $L(q)$ or $P(q)$ and each of its two ends fixes a point of the projective line. Then, up to projective conjugation, it has the displayed form in Theorem 11, with t generating the whole field.*

Proof. The two fixed points differ, since the whole group fixes no point. Their common generator B fixes both, so conjugate them to infinity and zero and write $B = \mathrm{diag}(t, 1)$. The first and last generators have the forms $A = \left(\begin{smallmatrix} \lambda & \mu \\ 0 & 1 \end{smallmatrix}\right)$ and $C = \left(\begin{smallmatrix} 1 & 0 \\ \nu & \kappa \end{smallmatrix}\right)$. The involution relations give $\lambda = -t^{-1}$, $\kappa = -t$ and $\mu\nu = 1 + t$. Here $t \neq 1$, since B is nonidentity, and $t \neq -1$: otherwise $\mu\nu = 0$ and the whole group fixes zero or infinity. Thus $\mu \neq 0$, and conjugation by $\mathrm{diag}(1, \mu)$ normalizes μ to one. This gives the displayed matrices. They all lie in $P(\mathbb{F}_p(t))$, so the group-order formulas force $\mathbb{F}_p(t) = F$. (For a proper subfield of size s, $q \geq s^2$ and $|L(q)| > |P(s)|$.) The preceding theorem applies. □

The full-field condition is $k \nmid p^e - 1$ for $0 < e < d$, where $q = p^d$; the absence of field reversal is $k \nmid p^e + 1$ for the same range. Thus this one theorem gives both source pointed-parabolic classifications, including their counts. For $P(q)$ with odd q, add $k \nmid (q-1)/2$; for $L(q)$ with odd q, add $k \mid (q-1)/2$ and k even. The latter implies $q \equiv 1 \pmod 4$. In characteristic two there is no determinant-square distinction and no even-k requirement.

There is a proof-route correction to the source's subfield sentence. When t generates F but a field automorphism inverts it, the group can still be all of $L(q)$ or $P(q)$. The exclusion from the chiral list comes from the semilinear reversal just constructed. For example, an element $t \in \mathbb{F}_{25}$ of order six gives $L(25)$, of order 7800, with affine ends of order 150 and intersection of order six; Frobenius inversion makes this marking regular. These orders follow uniformly from the theorem, not from a sample extrapolation.

9 Anchored involutions and field descent

The retained rotation correspondence uses generators $\sigma_1, \ldots, \sigma_{n-1}$ and the interval-involution relations. Its intersection axiom says that subgroups generated by interval products with endpoints in I and J intersect in the subgroup for $I \cap J$. We use this full axiom, not merely the rank-four special case.

Lemma 13. *In the rotation group of a rank-n chiral polytope, the involutions $t_i = \sigma_i \cdots \sigma_{n-1}$, $1 \leq i \leq n-2$, form a string C-marking of rank $n-2$. Its adjacent labels are $|\sigma_1|, \ldots, |\sigma_{n-3}|$, and its even subgroup $\langle \sigma_1, \ldots, \sigma_{n-3} \rangle$ has index two. There is a dual anchored marking with the labels at the other end.*

Proof. Every t_i is a nonidentity involution by the rotation correspondence. For $i < j$, cancellation gives $t_i t_j = \sigma_i \cdots \sigma_{j-1}$. If $j - i \geq 2$ this is an involution, proving the string commutations; if $j = i + 1$ it is σ_i. For a set of indices I, the group $\langle t_i : i \in I \rangle$ is exactly the endpoint subgroup of the rotation intersection axiom with vertex indices $\{i - 1 : i \in I\} \cup \{n - 1\}$. Products $t_i t_j$ supply every interval with both endpoints in this set. Intersecting two such sets therefore proves the complete C-intersection property, including the empty intersection. The even words in the t_i are generated by the adjacent products $\sigma_1, \ldots, \sigma_{n-3}$. This subgroup cannot contain t_{n-2}: its endpoint set meets that for t_{n-2} in a single vertex, whose endpoint subgroup is trivial. Thus its index is two. Reversing the indexing gives the dual statement. □

Lemma 14. *If a chiral rotation group is $L(q)$ or $P(q)$, $q \geq 4$, all its labels are greater than two.*

Proof. The first label cannot be two: the identity automorphism would then satisfy the rotation reversal rule $\sigma_1 \mapsto \sigma_1^{-1}$, $\sigma_2 \mapsto \sigma_1^2\sigma_2$ and fix the other generators. The same holds at the last label by duality. Suppose instead $\sigma_j^2 = 1$ with $1 < j < n-1$. Put $A = \langle \sigma_1, \dots, \sigma_{j-1} \rangle$ and $B = \langle \sigma_{j+1}, \dots, \sigma_{n-1} \rangle$. For $x = \sigma_i \cdots \sigma_{j-1}$ and $y = \sigma_{j+1} \cdots \sigma_k$, all of $x\sigma_j$, $\sigma_j y$ and $x\sigma_j y$ are involutions. Thus σ_j inverts each such x, y, and the last relation gives $xy = yx$. These prefix and suffix elements generate A, B. Hence A and B commute and are normal in $\langle A, B, \sigma_j \rangle = G$; both are nontrivial. This is impossible in $L(q)$ or $P(q)$: every nontrivial normal subgroup contains the simple socle $L(q)$, whose centralizer is trivial. These are exactly the standard structural inputs already retained in Topic E. No assumption about a minimal generating number is used. □

Lemma 15. *Let $k \subseteq K$ be finite fields of odd characteristic, $|k| \geq 5$. Any conjugate of $P(l)$ or $L(l)$ in $P(K)$ containing the standard $L(k)$ has $k \subseteq l$ and is the standard $P(l)$ or $L(l)$. Consequently its intersection with standard $P(k)$ is $P(k)$ in the first case. In the second case it is $P(k)$ if $[l : k]$ is even and $L(k)$ if $[l : k]$ is odd.*

Proof. The projective invariant $J(X) = \operatorname{tr}(X)^2/\det(X)$ lies in l on such a conjugate. The matrices $\left(\begin{smallmatrix} a & -1 \\ 1 & 0 \end{smallmatrix}\right) \in L(k)$ show that $a^2 \in l$ for every $a \in k$. Squares span k, since $a = ((a+1)/2)^2 - ((a-1)/2)^2$, so $k \subseteq l$.

Let ϕ be the $|l|$-power field automorphism of K. The standard $P(l)$ is its pointwise fixed projective group: normalize one nonzero matrix entry to one to check this assertion directly. A conjugate $P(l)^g$ is the fixed group of the conjugate semilinear automorphism θ. Both θ and ϕ fix $L(k)$ pointwise, so their quotient is an inner projectivity centralizing $L(k)$. That centralizer is trivial: it centralizes the upper and lower nonidentity elementary unipotents, hence fixes infinity and zero, and commutation with translation forces its diagonal ratio to be one. Thus $\theta = \phi$ and $P(l)^g = P(l)$. The same argument applied to an overgroup of $L(l)^g$ places it in standard $P(l)$; its simple index-two subgroup is standard $L(l)$. Finally a scalar in $k^\times$ is a square in l exactly when it is a square in k or $[l : k]$ is even. Applying this to determinants proves the intersections. □

We use the accepted Topic E classification: a projective group with a rank-four C-marking is $P(5)$, $L(11)$ or $L(19)$, with its specified marking. The proof and finite table belong to the complete common paper and remain paid once. We also retain Dickson's proper-subgroup alternatives in their natural-subfield form. Here are the precise additional interfaces.

Lemma 16. *A directly regular rank-three C-group contained in $P(q)$, with both labels greater than two, is either S_4 with its tetrahedral marking, or a natural conjugate of $P(k)$ with k odd, $|k| \geq 5$, whose rotation subgroup is $L(k)$. A directly regular rank-four C-group contained in $P(q)$, with all labels greater than two, is the simplex $P(5) \cong S_5$. No C-marking contained in $P(q)$ with these labels has rank at least five.*

Proof. In odd characteristic an affine subgroup has only half-turn involutions; two distinct such involutions cannot commute. It cannot contain a rank-three string marking. In characteristic two its involutions are translations and all commute. A cyclic group is likewise excluded. In a dihedral group, commuting endpoint involutions of a rank-three marking cannot be central, so are opposite reflections. Writing them $s, sr^{N/2}$ and the middle reflection sr^a, generation says $\gcd(a, N/2) = 1$. At least one of $a, a - N/2$ is coprime to N; the corresponding adjacent pair already generates the whole group, contradicting C-intersection. A central middle involution would instead give a label two.

All involutions in A_4 commute. The group A_5 has no index-two subgroup. A directly regular marking of S_4 consists of transpositions, since its even subgroup is its unique index-two subgroup A_4. Commuting distinct endpoints are disjoint transpositions and a middle transposition joining them gives precisely type $\{3, 3\}$. The remaining Dickson alternatives are natural projective subfield groups. The simple $L(k)$ has no index-two subgroup; $P(k)$ has one exactly in odd characteristic, namely $L(k)$. The small fields give the cases already handled. This proves the rank-three assertion.

For rank four, the affine and dihedral exclusions still apply to the first three generators. The involution centralizer in A_5 is a Klein four group, so a first generator commuting with the noncommuting last two is impossible. In S_4, if an endpoint were a transposition its centralizer would be abelian, giving the same contradiction. Thus both endpoints are double transpositions. The two middle generators must be transpositions, since adjacent labels exceed two. The first adjacent pair generates a dihedral group of order eight containing the entire normal Klein four group, hence the last generator, contrary to the intersection property. This also excludes A_4. Dickson now reduces to projective subfield groups. The common rank-four classification leaves $P(5), L(11), L(19)$; only $P(5)$ has an index-two subgroup, and its marking is the simplex marking.

Finally, at rank at least five both fourth and fifth generators would centralize the first noncommuting pair. The common projective centralizer lemma allows at most one nonidentity element there in odd characteristic; in characteristic two the centralizer of an involution is abelian and already prevents rank four. This is independent of which subgroup the marking generates. □

Theorem 17. *Neither $L(q)$ nor $P(q)$, $q \geq 4$, is the full group of a chiral polytope of rank at least five.*

Proof. All labels exceed two. Anchoring gives a directly regular C-subgroup of rank $n-2$, so $n \leq 6$. If $n = 6$, the anchored rank-four subgroup is a simplex, so the first three labels are three. Its dual makes the last three labels three. The full rotation group is therefore a quotient of the even simplex Coxeter group $[3,3,3,3,3]^+ \cong A_7$. This group is simple, so the nontrivial quotient is itself, with its regular simplex marking. This contradicts chirality.

For $n = 5$ write the rotations as $\alpha, \beta, \gamma, \delta$. The two anchored directly regular groups are

$$P_0 = \langle \alpha, \beta, \gamma\delta \rangle, \qquad Q_0 = \langle \alpha\beta, \gamma, \delta \rangle,$$

with even subgroups $R = \langle \alpha, \beta \rangle$ and $S = \langle \gamma, \delta \rangle$. If both are tetrahedral, all four labels are three and the same argument uses the simple $[3,3,3,3]^+ \cong A_6$ and its regular marking.

First suppose the whole group is $P(q)$ with q odd. The even group of either anchored subgroup lies in $L(q)$: it is a natural $L(k)$, or is A_4 generated by elements of order three, which must lie in the determinant-square kernel. Thus all four rotations lie in $L(q)$, a contradiction. If q is even there are no odd-characteristic natural $P(k)$ subgroups, so both anchored groups would be tetrahedral, already excluded.

It remains that the whole group is $L(q)$ with q odd, and at least one anchored group, say P_0, is standard $P(k)$, $|k| \geq 5$. Then $R = L(k)$ and $[\mathbb{F}_q : k]$ is even. Put $F_0 = \langle \alpha, \beta, \gamma \rangle$, a proper subgroup by rotation intersection. Dickson's alternatives containing R are projective subfield groups; the sole exceptional possibility is $R = F_0 = A_5$, which would imply $G = \langle P_0, \gamma \rangle = P_0$, impossible. The subfield-envelope lemma puts F_0 in standard $P(l)$ or $L(l)$ with $k \subseteq l$. The first possibility contains P_0, and again $G = \langle F_0, P_0 \rangle = F_0$ is impossible. In the second, the exact rotation intersection is $F_0 \cap P_0 = R = L(k)$. Thus $[l : k]$ is odd, while $[\mathbb{F}_q : k]$ is even. It follows that $[\mathbb{F}_q : l]$ is even, so $P(l)$ is a proper subgroup of $L(q)$. But $G = \langle F_0, P_0 \rangle \leq P(l)$, the final contradiction. □

10 A trace quadratic for an exceptional end

Throughout this section the field has odd characteristic. Traces refer to specified determinant-one lifts. Thus a projective element of order three has chosen trace 1, an element of order four has chosen trace $\sqrt{2}$, and the two classes of order-five elements have traces $(-1 \pm \sqrt{5})/2$. The last convention is used away from characteristic five. Changing the sign of a lift changes none of the projective statements.

Theorem 18. *Let F be a finite field of odd order, and let $a, b \in F^\times$, $c \in F$ with $d = a^2 + b^2 - 4 \neq 0$. Up to simultaneous projective conjugacy, pairs $A, B \in \mathrm{SL}_2(F)$ with traces a, b and $\mathrm{tr}(AB) = 0$ form one class. For each such pair, determinant-one matrices C with*

$$\mathrm{tr}(C) = c, \qquad \mathrm{tr}(BC) = \mathrm{tr}(ABC) = 0$$

are in bijection with the square roots in F of

$$\Delta = a^2c^2 - 4(a^2 + b^2 + c^2 - 4).$$

There are two, one or zero extensions according as Δ is a nonzero square, zero or a nonsquare. Their invariant is $2\operatorname{tr}(AC) - ac = \pm\sqrt{\Delta}$. If $a, c \neq 0$ and $c^2 \neq 4$, a projective conjugation realizes the rotation reversal precisely when $\Delta = 0$.

Proof. Put $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Choose $U, V \in F$ with $U^2 + V^2 = d$. Such a choice exists because the sets of squares and of d minus squares each have $(|F| + 1)/2$ elements. Set $u = (U + a)/2$, $w = (V - b)/2$ and

$$A = \begin{pmatrix} u & w + b \\ w & a - u \end{pmatrix}, \qquad B = A^{-1}J.$$

Direct expansion gives determinant one, the required traces, and $AB = J$. The nonzero d says that A, B have no common eigenline: equivalently, $\det(AB - BA) = -d$. More generally the four matrices I, A, B, AB are linearly independent. This follows by diagonalizing a nonscalar A: both off-diagonal entries of B are nonzero, so I, A span the diagonal matrices and B, AB supply the off-diagonal ones. In the remaining Jordan case its unique eigenline is not B-invariant, and the same four-entry calculation proves independence. The Cayley–Hamilton identities and $AB + BA = bA + aB - abI$ determine their multiplication table from a, b. Consequently any second pair with the same three traces defines an algebra automorphism of $M_2(F)$ by matching these bases. Such an automorphism is inner: take a nonzero vector in the image of E_{11}, and its image under the transformed E_{21}, as a basis. The transformed matrix units then act as the usual matrix units. This proves pair uniqueness.

Write $C = \left(\begin{smallmatrix} r & s \\ s & c-r \end{smallmatrix}\right)$, as is forced by $\operatorname{tr}(JC) = 0$, and put $R = 2r - c$, $S = 2s$. The other trace and determinant equations are exactly

$$VR - US = -bc, \qquad R^2 + S^2 = c^2 - 4.$$

Since $d \neq 0$, every solution is uniquely of the form

$$R = \frac{-bcV + zU}{d}, \qquad S = \frac{bcU + zV}{d}, \qquad z^2 = \Delta.$$

This also gives $2\operatorname{tr}(AC) - ac = UR + VS = z$. The projective centralizer of the ordered pair is trivial, so the extensions have not acquired an additional conjugacy identification.

For the last assertion, nonzero traces remove the signs in determinant-one lifts of $XAX^{-1} = A^{-1}$ and $XCX^{-1} = C$. Put $A_0 = A - aI/2$, $C_0 = C - cI/2$. The centralizer algebra of C is $F[I, C_0]$, so $X = xI + yC_0$. The first equation is

$$XA_0 + A_0X = 2xA_0 + y\operatorname{tr}(A_0C_0)I = 0.$$

Since A_0 is nonscalar, invertibility forces $x = 0$, $y \neq 0$ and $\operatorname{tr}(A_0C_0) = z/2 = 0$. Conversely, if $z = 0$, choose $X = C_0$. It is invertible because $c^2 \neq 4$, has scalar square, commutes with C, and inverts A projectively. Also $C_0J = -JC_0$ by the trace condition, so it fixes $AB = J$ projectively and sends B to A^2B. This is exactly the rotation reversal rule. □

Lemma 19. *If $q = p^d \equiv 3 \pmod 4$ and $d > 1$ is a prime power, then $L(q)$ cannot be a chiral rotation group.*

Proof. Only rank four remains. Neither end group fixes a point: it contains the involution given by the product of its rotations, whereas an involution of $L(q)$ has no rational fixed point when $q \equiv 3 \pmod 4$. By the retained natural-subfield maximal-subgroup classification, each proper end lies in a conjugate of $L(p^{e_i})$ with $e_i \mid d$ proper. Here the exceptional A_4, S_4, A_5 alternatives also lie in proper subfield overgroups; the odd degree excludes a natural $P(p^{e_i})$. Conjugate the first end to its standard position

and choose lifts A, B there. A lift of the third rotation has trace $c \in F_{p^{e_2}}$, since trace is conjugacy invariant. Put $F = F_{p^{\mathrm{lcm}(e_1,e_2)}}$; this is a proper subfield because the divisors of the prime power d are linearly ordered.

The first pair is absolutely irreducible. Otherwise its common algebraic eigenline is rational, contrary to the preceding fixed-point observation, or its Frobenius conjugate is a distinct common eigenline. In the latter case the pair is diagonal on the two lines and commutes. Then $(AB)^2 = 1$ makes $A^2 = B^{-2}$ projectively; the cyclic intersection axiom forces both rotations to have order at most two, already excluded. Thus $a^2 + b^2 - 4 \neq 0$. Normalize AB to J over F and apply the explicit extension formulas. All entries of C lie in $F(\sqrt{\Delta})$. The ambient extension has odd degree, so this quadratic extension is F itself. Hence all three rotations lie in the proper $L(F)$, a contradiction. This replaces the source's quadratic calculation, including its printed coefficient errors, by a nonsingular formula. □

11 Exceptional ends: three discriminants and the exact intersections

The finite input to this section is fully specified, rather than hidden behind a group-name lookup. The program `exceptional_amalgams.g` uses exact GAP subgroup and finitely presented group algorithms. It first enumerates every ordered generating pair (a, b) in A_4, S_4, A_5 with $|a|, |b| > 2$, $|ab| = 2$ and $\langle a \rangle \cap \langle b \rangle = 1$. The numbers of pairs are respectively $24, 48, 360$, and their types modulo automorphisms are

$$A_4 : (3,3), \qquad S_4 : (3,4), (4,3), \qquad A_5 : (3,5), (5,3), (5,5).$$

For every $(5,5)$ pair in A_5 it checks $ab^2a^3b^2a^2 = 1$. This is a complete finite premise, not a sample of projective groups. The standard spherical triangle presentations identify the $(3,3), (3,4), (3,5)$ groups as A_4, S_4, A_5.

Lemma 20. *If $p \equiv 3 \pmod 4$ is prime and $L(p)$ is a chiral rank-four rotation group, its type, up to duality, is $(5,3,4)$, $(5,3,5)$ or $(3,5,3)$.*

Proof. Neither proper end fixes a point. A dihedral end with both rotation orders greater than two is impossible: both rotations would lie in its cyclic rotation subgroup, and their trivial cyclic intersection together with $(ab)^2 = 1$ would force their orders to be at most two. The retained prime-field subgroup theorem therefore leaves A_4, S_4, A_5. The displayed complete local pair census leaves only the compatible three-label types. The types $(3,3,3)$, $(3,3,4)$, $(3,4,3)$ and $(3,3,5)$ and their duals are excluded by their spherical universal rotation groups. These have respectively order $60, 192, 576, 7200$; the latter has only C_2 and A_5 composition factors, and the middle two are solvable. No simple quotient is $L(p)$ with $p \equiv 3 \pmod 4$. These are the standard simplex, cube, 24-cell and 120-cell groups; their full presentation orders and composition factors are also checked by the same finite program.

For $(4,3,4)$ the trace quadratic has $a^2 = c^2 = 2$, $b^2 = 1$ and $\Delta = 0$. Its explicit projective reversal excludes chirality. The remaining unwanted types are $(5,5,3)$ and $(5,5,5)$ and their duals. Add the preceding five-five relator to each five-five end of its full rotation presentation. The exact finite calculation gives order 7200 and a composition series with successive orders $7200, 120, 2, 1$ in both cases. Their only nonabelian simple composition factor is A_5. Again $L(p)$ cannot be a quotient. This exhausts every compatible local pair, without an unrecorded search in $L(p)$. □

For completeness we give the full finite intersection test. Write

$$T(r,s,t) = \langle a, b, c \mid a^r = b^s = c^t = (ab)^2 = (bc)^2 = (abc)^2 = 1 \rangle.$$

In an A_5 end of type $(5,3)$ or $(3,5)$ the word $u(a,b) = ab^2a^2b^2a^2$ is an involution inverting b and generates, with b, its full dihedral normalizer. In an S_4 end of type $(3,4)$ the corresponding word is $v(c,b) = cb^2c^2$. All these word assertions are checked on every relevant local pair. The only proper overgroups of $\langle b \rangle$ in A_5 have orders $3, 6, 12$ when $|b| = 3$, and $5, 10$ when $|b| = 5$. Thus a dihedral

collision between the two ends forces $u(c,b) = u(a,b)b^k$ (using v in the S_4 case). For $k = 1, \dots, s$ the exact quotient orders are

(r,s,t)	$\lvert T(r,s,t)/\langle\langle \text{collision} \rangle\rangle \rvert$
$(5,3,4)$	$60, 1, 1$
$(5,3,5)$	$3420, 60, 60$
$(3,5,3)$	$660, 1, 360, 360, 1.$

For an A_4 collision put $u_1 = a^2b^{-1}a$, $u_2 = ab^{-1}a^2$. The two subgroups $\langle u_i, b\rangle$ are exactly the A_4 overgroups in the first A_5. Their three involutions are the b-conjugates of $u_i b$. At the other end use $v = b^2c^2$ for S_4, or $v_1 = c^2b^{-1}c$, $v_2 = cb^{-1}c^2$ for A_5. It is therefore enough, and is exhaustive, to add each relation $u_i^{b^j} = v_l$, $j = 0, 1, 2$. The six $(5,3,4)$ quotients all have order two. For $(5,3,5)$ the twelve orders, in lexicographic (i,l,j) order, are

$$1,1,1, \quad 1,60,60, \quad 60,1,60, \quad 1,1,1.$$

This corrects the source's assertion in Lemma 5.12 that all twelve quotients are trivial. Its exclusion of A_4 intersections outside $L(4) \cong L(5)$ remains valid. Full finitely presented group orders, not orders of selected permutation images, are used in this calculation. The nontrivial collision groups are identified as A_5, $L(9)$, $L(11)$ and $L(19)$ at orders $60, 360, 660, 3420$. Their simplicity shows that the complete intersection conclusions hold for arbitrary projective quotients: exclude $q = 4, 5$ for $(5,3,4)$; $q = 4, 5, 19$ for $(5,3,5)$; and $q = 9, 11$ for $(3,5,3)$. Thus the source's three intersection lemmas, not merely their prime-field applications, follow from this one finite interface.

Theorem 21. *Let $p \geq 7$ be prime. For each square root of 5 in F_p define*

$$D_{534} = 1 \pm \sqrt{5}, \qquad D_{535} = (7 \pm 5\sqrt{5})/2, \qquad D_{353} = 3 \pm 2\sqrt{5}.$$

For type $(5,3,4)$ also require that 2 is a square in F_p. For any of these three types, the number of ordered chiral polytope isomorphism classes with rotation group $L(p)$ is twice the number of nonzero squares in the corresponding two-element list. In the $(3,5,3)$ case this formula gives zero at $p = 11$; in the $(5,3,5)$ case it gives two at $p = 19$. A missing square root of 5 means there are no such classes. The same types with group $L(q)$, $q \equiv 3 \pmod 4$, require $q = p$ prime.

Proof. A local $(5,3)$ or $(5,5)$ pair generates A_5 by the spherical triangle presentation and simplicity. A $(3,4)$ pair generates S_4: the exact order four prevents its proper triangle quotients. Fix traces as above. The nonzero quantity $d = a^2 + b^2 - 4$ follows either from the absolute irreducibility of the A_5 end or directly from the quadratic equation for its order-five trace. The trace extension theorem gives all extensions and their conjugacy classes. For $(5,3,4)$ it gives D_{534}, while for $(3,5,3)$ it gives D_{353}. For $(5,3,5)$, if $a^2 = c^2$, it gives D_{535}. If the two order-five trace squares differ, their sum is three and product one, so $\Delta = 1$. Both resulting extensions lie inside the first A_5. Indeed, for the fixed pair $a = (1\,2\,3\,4\,5)$, $b = (2\,5\,3)$ the complete local enumeration finds exactly two order-five elements c satisfying both involution relations, and both are in the other A_5 conjugacy class from a. The two classes have different trace squares in every characteristic under consideration: replacing an order-five element by its square replaces its trace square t by $(t-2)^2 = 3 - t$, using $t^2 - 3t + 1 = 0$; these differ because the characteristic is not five. Pair uniqueness and the two-root formula show that these are exactly the mixed-class extensions. They cannot generate $L(p)$.

Every remaining extension generates $L(p)$. A subgroup containing A_5 in a prime-field $L(p)$ is either that A_5 or the whole group, by the retained subgroup classification. In type $(5,3,4)$, c has order four and hence is outside A_5. In type $(5,3,5)$ the preceding complete local check excludes same-class c inside A_5. In type $(3,5,3)$ the complete local check finds no order-three c inside the first A_5 satisfying the two involution relations. Thus generation is uniform; no large-group enumeration is substituted for it.

The intersection table proves the end intersection is exactly $\langle b \rangle$, except possibly at $p = 19$ in type $(5, 3, 5)$ and $p = 11$ in type $(3, 5, 3)$. Adjacent cyclic intersections are trivial by the coprime orders. The required program `exceptional_small.g` checks every root of the same equations at these two primes. At $p = 19$ the two nonzero-root extensions have end intersection of order three; the zero-root extension has intersection six. At $p = 11$ the sole $(3, 5, 3)$ extension has zero discriminant and intersection ten. Thus every nonzero-root extension has the full rank-four rotation intersection property, and none of the zero-root cases contributes.

For a prime field every automorphism of $L(p)$ is induced by a projectivity, a retained classical input. The trace extension theorem therefore rules out reversal for every nonzero root. It produces the reversal at a zero root whether or not the intersection property holds. The two choices of order-five trace square cannot be identified by an automorphism, and the invariant $2\,\mathrm{tr}(AC) - ac$ distinguishes the two roots within a choice. Conversely pair uniqueness shows that these are all classes. This proves the exact counts.

Finally suppose $q = p^d \equiv 3 \pmod 4$ and such a type is realized. Here d is odd. Its local A_5 (and S_4 where applicable) traces lie in an extension of the prime field of degree at most two, so they already lie in F_p. The same applies to a square root of Δ inside F_q. Conjugate the first pair to the displayed prime-field normal form. The unique extension formulas put the third matrix in $\mathrm{SL}_2(p)$ as well. Hence the generated group is conjugate to $L(p)$, and equality with $L(q)$ forces $d = 1$. □

The residue conditions of the source now follow from ordinary quadratic reciprocity. The products of the paired discriminants are respectively $-4, -19, -11$. For $p \neq 19, 11$ the Legendre symbols of the latter two are $(p/19)$ and $(p/11)$. The quadratic residues modulo 19 are $1, 4, 5, 6, 7, 9, 11, 16, 17$; those modulo 11 are $1, 3, 4, 5, 9$. If the product is a nonsquare exactly one discriminant is a square; if it is a square both are squares or neither is. For $(5, 3, 4)$ the order-five and order-four trace requirements give $p \equiv 1, 9, 31, 39 \pmod{40}$. At residues $31, 39$ exactly one D_{534} is a square, and at $1, 9$ either both or neither are. These are the complete counts of source Theorems 6, 7, 9 and 11. In particular residue $10 \pmod{11}$ belongs to the nonsquare-product case of Theorem 11. Its omission from Theorem 1(4b) in the exact arXiv-v1 edition is a statement error, not an additional exception to the construction. A separately recorded $p = 571$ matrix witness makes that omission explicit.

Proposition 22. *For $q \geq 4$, $P(q)$ occurs as the full group of a chiral rank-four polytope exactly when $q > 4$. For $q \equiv 1 \pmod 4$, $L(q)$ occurs exactly when $q > 9$.*

Proof. For $P(q)$ choose a primitive t, of order $q - 1$. If $q > 4$, its order exceeds $p^{d-1} + 1$ when $d > 1$, so no nonidentity field automorphism fixes or inverts it. The pointed theorem applies, with nonsquare t in odd characteristic. For $L(q)$, $q \equiv 1 \pmod 4$, choose t of order $(q - 1)/2$. This is even and greater than two when $q > 9$. For $d > 1$ it exceeds $p^{d-1} + 1$ except at $p^d = 9$: the required strict inequality is $p^{d-1}(p - 2) > 3$. Again the pointed theorem applies. At $d = 1$ there is no field automorphism to test.

The omitted groups are $L(4) = P(4) = L(5) \cong A_5$ and $L(9) \cong A_6$. The complete finite proof in `small_rotation_exclusions.g` examines all pairs (a, b) modulo the actual automorphism group, retaining $|a|, |b| > 2$, $|ab| = 2$, trivial cyclic intersection and a proper first end. For each it examines every $c = (ab)t$ with t an involution; this is exactly the condition $(abc)^2 = 1$. It then applies the remaining relations, generation and complete rank-four intersection conditions. There are one and seven pair representatives, respectively. The resulting polytopal triples number one and four (before the final stabilizer quotient), and each has a verified involutory automorphism $(a, b, c) \mapsto (a^{-1}, a^2 b, c)$. Thus none is chiral. This is a complete finite premise for these groups, separate from the uniform existence proof. □

12 Primitive squares in two odd extensions

This section reproduces the project's earlier accepted primitive-square proof, independently audited in phase267. Its new placement in the common theory is not a new answer to a source question. The

retained analytic input is the multiplicative-character bound in Cohen–Kapetanakis–Reis, *The existence of F_q-primitive points on curves using freeness*, Theorem 5 (printed p. 643), together with the cyclic-group primitive-element indicator. The exact primary and prior proof bindings accompany the scope map. The following particular norm construction and finite reduction are the previously proved work.

Theorem 23. *Let p be an odd prime and $e_1, e_2 > 1$ odd integers. There are primitive $j_i \in F_{p^{e_i}}$ such that, with $u = j_1 + j_1^{-1}$ and $v = j_2 + j_2^{-1}$,*

$$u^2v^2 - 4(u^2 + v^2)$$

is a nonzero square in $F_{p^{\mathrm{lcm}(e_1,e_2)}}$. The degrees may be equal or nested.

Proof. Reorder the degrees and put $g = \gcd(e_1, e_2)$, $q_0 = p^g$, $a = e_1/g \le b = e_2/g$. Then a, b are coprime and odd. Write $E = F_{q_0^a}$, $K = F_{q_0^b}$, $L = F_{q_0^{ab}}$. Fix any primitive $j_1 \in E$ and set $s = (j_1 + j_1^{-1})^2$. One has $s \neq 0, 4$, since otherwise j_1 has order at most four. Also s has degree a over F_{q_0}: if it lies in a proper intermediate field, the equations $u^2 = s$ and $j_1^2 - uj_1 + 1 = 0$ put first u and then j_1 in that same field, because the remaining extension degree is odd. This contradicts primitiveness.

Define

$$f_s(X) = (s-4)(X^2+1)^2 - 4sX^2, \qquad P(X) = \prod_{i=0}^{a-1} f_{s^{q_0^i}}(X).$$

For $x \neq 0$ the desired quantity with $j_2 = x$ is $f_s(x)/x^2$. The quadratic in X^2 defining f_s has nonzero constant and leading coefficients $s - 4$ and discriminant $64s \neq 0$. Thus f_s has four distinct nonzero roots. Distinct quartics have disjoint roots: subtracting gives $(s-t)(X^2-1)^2$, while $f_s(\pm 1) = -16 \neq 0$. Consequently $P \in F_{q_0}[X]$ is squarefree of degree $D = 4a$ and $P(0) \neq 0$. Since a, b are coprime, the q_0^b-Frobenius permutes all these a conjugates. For $x \in K$ this proves

$$N_{L/K}(f_s(x)) = P(x), \qquad \chi_L(f_s(x)/x^2) = \chi_K(P(x)),$$

where χ is the quadratic character extended by zero. If $a > 1$, P has no root in K: a root would give $s^{q_0^i} = 4(x^2+1)^2/(x^2-1)^2 \in E \cap K = F_{q_0}$, contradicting its degree. When $a = 1$, at most four roots remain.

Put $Q = |K|$, $n = Q - 1$, $W(n) = 2^{\omega(n)}$ and $\vartheta(n) = \varphi(n)/n$. For every multiplicative character ψ,

$$\left| \sum_{x \in K^\times} \psi(x)\chi_K(P(x)) \right| \le D\sqrt{Q}. \tag{20}$$

Indeed, for nontrivial $\psi = \eta^k$, $1 \le k < n$, with η a generator of the character group, the summand is $\eta(x^k P(x)^{n/2})$. Its polynomial has $D + 1$ distinct roots and at a root of P has multiplicity exactly $n/2$, so is not a scalar times an nth power. The retained bound gives $D\sqrt{Q}$. For trivial ψ, apply the quadratic-character bound on all of K and remove zero, giving $(D-1)\sqrt{Q}+1 \le D\sqrt{Q}$. All characters, including the trivial one, are extended by zero at zero.

The primitive-element indicator is

$$\mathbf{1}_{\mathrm{primitive}}(x) = \vartheta(n) \sum_{d \mid n} \frac{\mu(d)}{\varphi(d)} \sum_{\mathrm{ord}\,\psi = d} \psi(x).$$

Character orthogonality in the cyclic group $K^\times$ proves this formula. There are $\varphi(d)$ characters of order d and $W(n)$ squarefree divisors. If Z counts primitive roots of P, (20) therefore makes the number of primitive x with $P(x)$ a nonzero square positive provided

$$Q - 1 > DW(Q-1)\sqrt{Q} + Z/\vartheta(Q-1).$$

For $a > 1$ it suffices that

$$Q - 1 > 4aW(Q-1)\sqrt{Q}; \tag{21}$$

for $a = 1$, since $Z \le 4$ and $1/\vartheta(n) \le W(n)$, it suffices that

$$Q - 1 > 4W(Q-1)(\sqrt{Q} + 1). \tag{22}$$

We give the full finite reduction. For every positive integer n,

$$W(n) \le \frac{64}{30030^{1/4}} n^{1/4} < 5n^{1/4}.$$

Each prime at least 17 contributes $2 \le \ell^{1/4}$; only $2, 3, 5, 7, 11, 13$, whose product is 30030, can enlarge the ratio. Repeated prime powers decrease it, and $64^4 < 5^4 \cdot 30030$. Thus (21) holds when $Q > (21a)^4$. In the nonnested case $3 \le a \le b - 2$, $b \ge 5$. For odd $b \ge 23$, $3^b > [21(b-2)]^4$, first by direct comparison at 23 and then because the ratio increases on replacing b by $b + 2$ by a factor at least $9(21/23)^4 > 1$. For odd $5 \le b \le 21$, it remains only to examine $q_0^b \le [21(b-2)]^4$ with $q_0 = p^g$, g odd. The possible bases are at most $27, 13, 9, 5, 5, 3, 3, 3, 3$ in this order; the only composite base of odd prime-power exponent is 27. The exact integer comparison

$$(Q-1)^2 > 16a^2W(Q-1)^2Q$$

for all 66 remaining parameter triples leaves precisely

$$(q_0, a, b) = (3,3,5), (5,3,5), (3,3,7), (3,5,7), (3,5,9), (3,7,9).$$

This exhaustive bounded arithmetic is part of the required program, using exact trial division and integer comparisons.

For $a = 1$, (22) follows from $Q > 21^4$: writing $t = Q^{1/4} > 21$, $(t-20)t^3 > t^3 > 20t + 1$ gives $Q - 1 > 20t^3 + 20t$. There are exactly 23 odd prime-power fields $p^e \le 21^4$, $e \ge 3$ odd. The exact tests $Q - 1 - 4W > 0$ and $(Q - 1 - 4W)^2 > 16W^2Q$ leave only $Q = 3^3, 3^5, 5^3, 7^3, 11^3$. Their exponents are prime, so the divisor $e_1 > 1$ of e_2 equals e_2.

Finally put $z_e = Z(p^e)$ in the bound Conway-field representation, choose $j_1 = z_{e_1}$ and $j_2 = z_{e_2}^k$, and use the following complete exceptional list. The displayed exponent is even and nonzero modulo $p^{e_2} - 1$, and k is coprime to that integer.

p	e_1	e_2	k	$P(j_2)$ or $f_s(j_2)$
3	3	5	7	z_5^{228}
5	3	5	1	z_5^{2802}
3	3	7	1	z_7^{1336}
3	5	7	1	z_7^{406}
3	5	9	5	z_9^{18778}
3	7	9	1	z_9^{17426}
3	3	3	1	z_3^6
3	5	5	3	z_5^{200}
5	3	3	1	z_3^{84}
7	3	3	1	z_3^6
11	3	3	3	z_3^{906}

The complete polynomial coefficients and defining irreducible polynomials are carried in `psl2_square_witnesses.json`. The required `verify_square_witnesses.py` checks the finite-field arithmetic in explicit polynomial quotient rings, independently of GAP; the required `psl2_square_bound_check.py` checks both bounded reductions. The old GAP witness generators are retained separately as validation. These eleven exact certificates close all exceptions to the uniform inequality and hence all parameters of the theorem. □

13 Separating two subfield ends

The following is the complete previously proved intersection argument, now placed next to the common trace obstruction it uses. Its earlier independent acceptance and finite diagnostic are provenance, not an unpriced substitute for the proof below.

Lemma 24. *Let F_1, F_2 be subfields of a field K, with $F_1 \cap F_2 = F$. If $t \in K \setminus (F_1 + F_2)$, then*

$$L(F_1) \cap U(t)L(F_2)U(-t) = \{[U(b)] : b \in F\}, \qquad U(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix},$$

in every characteristic.

Proof. Write a determinant-one representative in the intersection as $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ over F_1. Its representative over the other field differs only by a scalar ± 1, which can be absorbed. Thus

$$U(-t)MU(t) = \begin{pmatrix} a - tc & b + t(a-d) - t^2c \\ c & d + tc \end{pmatrix}$$

has entries in F_2. Hence $c \in F$. If $c \neq 0$, the first entry would put t in $F_1 + F_2$, so $c = 0$. Now $a, d \in F$; if $a \neq d$, the upper-right entry gives the same contradiction. Thus $a = d = \pm 1$ and $b \in F$. Conversely every displayed unipotent belongs to both groups and commutes with $U(t)$. □

Theorem 25. *Let p be odd and $e_1, e_2 > 1$ coprime odd integers. Put $F_i = F_{p^{e_i}}$, $K = F_{p^{e_1 e_2}}$. Choose primitive $j_i \in F_i$ by Theorem 23, put $u = j_1 + j_1^{-1}$, $v = j_2 + j_2^{-1}$ and choose $D \neq 0$ with $D^2 = u^2v^2 - 4(u^2 + v^2)$. The projective matrices*

$$\alpha = \frac{1}{2}\begin{pmatrix} u & u+2 \\ u-2 & u \end{pmatrix}, \quad \beta = \frac{1}{2}\begin{pmatrix} 2+u & u \\ -u & 2-u \end{pmatrix}, \quad \gamma = \frac{1}{2u}\begin{pmatrix} v(u-2) & D \\ D & v(u+2) \end{pmatrix}$$

give a chiral rank-four polytope with full group $L(K)$ and type $((p^{e_1}-1)/2, p, (p^{e_2}-1)/2)$.

Proof. The odd-degree argument used above shows that u^2 and v^2 generate their full respective fields over F_p; in particular $u, v \neq 0$ and $u^2, v^2 \neq 4$. The matrices have determinant one, with traces $u, 2, v$. Their eigenvalues give the first and last orders, while $\beta = I + N$ with $N \neq 0$, $N^2 = 0$, gives order p. These orders imply the adjacent cyclic intersections are trivial. Direct multiplication gives $\alpha\beta = J$ and zero traces for $\beta\gamma, \alpha\beta\gamma$, so all interval-involution relations hold.

We prove the remaining intersection without assuming generation. If $x \in F_1$, $y \in F_2$ and $xy \in F_1 + F_2$, then $x \in F_p$ or $y \in F_p$. Indeed, apply the p^{e_1}-Frobenius τ to $xy = a + b$, $a \in F_1, b \in F_2$, and subtract. If $y^\tau \neq y$, then $x = (b^\tau - b)/(y^\tau - y) \in F_2$, hence in F_p; otherwise $y \in F_p$. Put $z = D/(uv)$. It satisfies $z^2 = 1 - 4/u^2 - 4/v^2$ and does not belong to $F_1 + F_2$. For if $z = x + y$ with $x \in F_1, y \in F_2$, the square equation puts $2xy$ in that sum. The preceding observation forces z into one of the two fields, and then puts u^2 or v^2 in F_p, a contradiction.

With $M^T = T^{-1}MT$, put

$$T = \begin{pmatrix} 1 & 2/u \\ -1 & 0 \end{pmatrix}, \qquad A(w) = \begin{pmatrix} 1 & (2-w)/w \\ -w & w-1 \end{pmatrix}, \qquad t = \frac{D}{2uv} + \frac{1}{v} - \frac{1}{2} - \frac{1}{u}.$$

Direct multiplication gives

$$\beta^T = U(1), \quad \alpha^T = A(u), \qquad (\gamma^T)^{U(t)} = A(v),$$

$$\gamma^T = \begin{pmatrix} (-D + v(u+2))/(2u) & -D/u^2 \\ -v & (D + v(u-2))/(2u) \end{pmatrix}.$$

The last identity also follows from its upper-left entry 1, lower-left entry $-v$, trace v and determinant one. Since $t \notin F_1 + F_2$, Lemma 24 puts the two actual end groups' intersection inside $\langle U(1) \rangle$. Both contain that group, so their intersection is exactly $\langle \beta \rangle$.

For generation use the retained maximal-subgroup list. The unipotent β^T fixes only infinity, whereas $A(u)$ does not fix infinity, so the group is in no point stabilizer. The dihedral maximal orders $|K| \pm 1$ are prime to p and cannot contain β. The exceptional A_4, S_4, A_5 have no element of order at least 13, whereas $|\alpha| = (p^{e_1} - 1)/2 \geq 13$. Finally the conjugacy invariant $\mathrm{tr}(M)^2/\det(M)$ takes the values u^2, v^2 on α, γ. A conjugate subfield group containing them must therefore contain both F_1, F_2 in its field of definition, and hence all of K. No proper maximal subgroup remains.

A reversing automorphism is semilinear by the retained automorphism theorem. Its field part fixes u^2 and v^2, so fixes K and is trivial. The trace obstruction in Theorem 18, or its final two-line centralizer calculation alone, now applies:

$$\mathrm{tr}(\alpha\gamma) - uv/2 = D/2 \neq 0.$$

Thus no projectivity can simultaneously invert α and fix γ. There is no reversal, so the full rotation correspondence gives the asserted chiral polytope. □

Every odd integer $d > 1$ with at least two distinct prime divisors admits a factorization $d = e_1 e_2$ into coprime odd integers greater than one, by separating its prime-power factors. This theorem supplies all such exponents. Together with the earlier exclusions, the nonprime-field classification is consequently: $L(p^d)$, $d > 1$, occurs in rank four exactly for even $q > 4$, for $q \equiv 1 \pmod 4$ with $q > 9$, or for $q \equiv 3 \pmod 4$ when d is not a prime power. This reproduces the previously accepted Section 6 completion. It is kept separate from the source's established principal theorem and earns no new question or historical-priority claim in this edition.

14 Attribution and computational boundary

The original three applications retain Leemans–Schulte's two projective polytope classifications and Leemans–Toledo's maniplex results. Their complete local and universal table, geometry and literal correction witnesses remain in `required_finite.g`. The rotation application recovers Elliott–Leemans–O'Reilly-Regueiro and Leemans–Moerenhout– O'Reilly-Regueiro, with edition-specific corrections made explicit. The primitive-square and coprime-subfield completion reproduces the project's earlier accepted work; it is not new question credit.

The new required programs are the complete exceptional amalgam and small prime certificates, the exhaustive A_5/A_6 exclusions, and the unchanged primitive-square integer reduction and polynomial witness verifier with its full JSON data. All whole files accompany the common paper. Finite samples of the pointed construction, the explicit 571 source correction, the earlier GAP witness generators and the subfield-intersection example are validation, with their whole files disclosed separately. No sampled large projective group supplies a uniform theorem.

Nine new Lean declarations prove the actual 2×2 intertwining identities and the bijective algebra behind the trace quadratic. Existing common normalizer, matrix and extension certificates, six primitive-square polynomial declarations and the actual subfield-entry statements are retained with their exact scopes. Irreducibility and finite-field existence, Dickson's theorem, projective automorphisms, the character bound, GAP's finite-presentation computations and the geometric correspondence remain written or retained inputs. No full Lean proof of a source is asserted. The source map

distinguishes the principal contracts from quoted all-field Coxeter-quotient classifications and motivating atlas tables not regenerated here. Full external proof and catalogue costs are unknown, not zero.

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